

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5.	(a)			Any 2 pairs of marks: Less energy is used to produce bioplastic (1) So less carbon footprint to produce electricity / less impact on global warming / climate change / greenhouse effect (1) does not rely on crude oil / made from plants <u>not oil</u> / does not rely on fossil fuel (1) renewable /crude oil is limited/less CO₂ produced (1) biodegradable (1) so less land fill / can be used as {biofuels/compost} (1) Not litter: / plastic waste requires less water (1) saves natural resources/less processing of water (1) smaller carbon footprint (1) less impact on global warming / climate change / greenhouse effect (1) less landfill (1) Less habitats destroyed / less land used (1)	2	2		4		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			(Land is used to grow plants to make bioplastic) Encourages deforestation reduce habitats / reduces biodiversity (1) OR reduce habitats (1) reduces biodiversity (1) OR Land not used for food production (1) Decreased availability to food / may lead to food shortages (1) OR When it biodegrades it produces methane, carbon dioxide and water / greenhouse gases (1) Which add to greenhouse effect / global warming / climate change (1) OR energy is used to produce bioplastic (1) So impact on global warming / climate change / greenhouse effect / carbon footprint (1) OR requires water (1) uses natural resources/processing of water (1) OR has a carbon footprint (1) impact on global warming / climate change / greenhouse effect (1) OR uses landfill (1) habitats destroyed / less land used (1)		2		2		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			How much CO ₂ equivalent is absorbed by the plant as it is grown (1) Needs to be at least 0.5 gCO ₂ Eq / kg (1)		1	1	2		
				Question 5 total	2	5	1	8	0	0